PhytoSense AI — Complete Product & Prototype Dossier

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Executive summary

PhytoSense AI builds **PhytoHydra**, a stem-mounted sensor node that measures how water is moving *inside* a living plant, so growers can act on hydraulic stress before leaves show it. It is a TRL 3 concept today; the 12-month goal is a field-validated TRL 5 prototype on arecanut in Karnataka.

The one-line claim. Soil probes, weather stations and cameras measure the plant's surroundings or its late symptoms. PhytoHydra measures the trunk itself, at several heights, with three independent signals, and turns them into a plant-specific hydraulic state score.

How it works, in one paragraph. A flexible band carries 4 electrode rings up the stem. The node injects a tiny, safe AC current (1 kHz to 1 MHz) and measures the voltage between ring pairs. This gives the stem's electrical impedance per segment and the *transfer function* between segments. Water content and xylem conduction change these numbers. An ultrasonic acoustic-emission sensor listens for cavitation clicks (air bubbles breaking water columns under tension). A micro-dendrometer records daily stem shrink and swell. Air temperature, humidity, light and soil moisture give context. An on-device model learns each plant's healthy baseline and reports deviations, a 0 to 100 hydraulic stress index, a location estimate and a 24 to 48 h trend.

What is new. Each of these sensing methods exists in research labs. Nobody sells them fused, distributed along one stem, with a per-plant baseline, at a price an Indian plantation can pay. That fusion, the electrode interface for palms, and the dataset are the defensible assets.

What we are asking for. A GENESIS EIR grant to fund the hardware, controlled-stress experiments on 24 arecanut seedlings, 10 field nodes in one plantation, and AI model development (fund plan in section 14).

What must be proven first. That the fused signal detects water stress earlier than soil moisture alone, measured against leaf water potential (pressure chamber) as ground truth. Every lead-time and savings number in the current deck is a hypothesis until that

experiment is done; this dossier labels them that way.

1. The problem

Growers irrigate by calendar and soil feel, not by what the plant needs, so they over-water in good weeks and react too late in bad ones. In arecanut the stakes are high: a mature garden yields a crop worth ₹3 to 5 lakh per tonne at Shivamogga APMC (modal prices of ₹39,000 to ₹54,000 per quintal in mid-2026, acrop.app).

Four concrete failures of today's practice

1. **Wrong signal.** Soil moisture says how wet one spot of soil is. It does not say whether the palm's roots are taking that water up, or how much tension its xylem is under at 2 pm on a hot day.
2. **Fixed schedules.** Extension advice is a flat rule: about 175 litres per palm every 4 to 7 days by hose, or 16 to 20 litres per palm per day by drip (DASD, ICAR-CPCRI). Real need varies with palm age, soil, shade and weather.
3. **Late symptoms.** Yellowing, frond drooping and nut drop show up days to weeks after stress began inside the plant. Cameras and satellites see these late symptoms.
4. **Water is getting scarce where arecanut is expanding.** Arecanut area in Karnataka roughly doubled between 2017 and 2021, much of it in drier plateau districts where growers drill borewells 1,200 to 1,600 ft deep (Krishi Jagran).

Why existing tools fall short

Tool	What it measures	What it misses
Soil moisture probe	Water in one soil volume	Root uptake, plant tension, variation between palms
Weather station + ET model	Atmospheric demand	The individual palm's response
RGB / thermal / satellite imaging	Canopy colour and temperature	Early, internal stress; hard under dense palm canopy
Single-point plant electrophysiology (e.g. Vivent)	Electrical potentials at one point	Where in the plant, hydraulic meaning; mainly greenhouse crops
Dendrometer service (e.g. Phytech)	Trunk shrink and swell	Cause of change; palms shrink very little

Tool	What it measures	What it misses
Stem water potential probe (e.g. Saturas)	Xylem tension at one drilled point	Requires drilling; one point; imported cost

The gap PhytoSense targets: **a minimally invasive (no drilled probe), multi-point, multi-signal view of water movement inside one palm, priced for Indian plantations.**

2. The core idea and the science behind it

The idea: water moving through a stem changes three things we can measure from the outside without drilling. Its electrical impedance, the ultrasonic clicks it makes when water columns break, and the stem's tiny daily shrink-and-swell. Measured at several heights on one palm and compared against that palm's own healthy baseline, these become an early, local stress signal.

Signal 1: Stem electrical impedance and the transfer function (primary)

- Living sapwood conducts electricity mainly through its water and ions. When the stem dehydrates, resistance rises. Lab work on six tree species found stem resistivity tracked tree water status linearly as trees dried (Ganthaler et al. 2019, Frontiers in Plant Science). Lysimeter work on mango, banana, date palm and olive found stem water content reacts to water stress within minutes (Nadler et al. 2008). A multi-electrode study on olive found daily relative resistance change rose with drought stress (Agricultural Water Management 2020).
- Measuring at several frequencies (impedance spectroscopy) separates water outside cells (low-frequency current goes around cells) from total water (high-frequency current passes through membranes). The ratio is a cleaner hydration index than one resistance value.
- PhytoHydra's twist: 4 electrode rings at different heights. Injecting current at one ring and reading voltage at the others gives the **transfer function** H(f) between stem segments, i.e. how the signal attenuates and phase-shifts along the stem. A drop in one segment's conduction shows up as a local change, which is how we *localise* a problem.

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H_{ij}(f,t) = \frac{V_j(f,t)}{V_i(f,t)}, \qquad Z_{seg}(f,t) = \frac{V_{seg}(f,t)}{I_{inj}(f,t)}
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Signal 2: Ultrasonic acoustic emission (validator)

- When xylem tension gets too high, water columns snap (cavitation) and release a short ultrasonic pulse. Piezo sensors on the trunk pick these up in the 100 kHz to 1 MHz band; in grapevine the method detected moderate drought stress at leaf water potentials of −0.7 to −1.2 MPa (Oletić et al. 2020).
- Not every click is cavitation. Frequency clustering separates cavitation-related signals (strong 100 to 200 kHz content) from others (Vergeynst et al. 2016, Tree Physiology). We use AE as a corroborating event count, not a standalone signal.

Signal 3: Micro-dendrometry (sanity check)

- Stems shrink by day as water is pulled out and refill at night. The daily shrinkage amplitude and the overnight recovery are classic plant-water indicators; Phytech's commercial service is built on them (AgInfo). A cheap strain-gauge band gives us this third, independent view.

Context signals: air temperature, humidity (to compute vapour pressure deficit, VPD), light (PAR), soil moisture at 30 cm, and stem temperature. Stem temperature matters: resistivity changes strongly with temperature and must be corrected (Ganthaler et al. 2019).

Palm-specific caveats we must design for

- Arecanut is a palm (monocot). It has no bark-and-cambium ring and no secondary xylem like a mango tree. Its vascular bundles are scattered through the whole trunk, and the outer trunk is hard and fibrous. Electrode contact and current paths will behave differently from the tree studies above. This is the single biggest technical unknown.
- Palms change diameter very little once mature, so dendrometer signals will be small (micrometres). Use it as a secondary signal.
- Most published AE and impedance work is on woody dicots. Arecanut-specific evidence does not yet exist; generating it is the core of the PoC and a publishable result in itself.

The honest scientific claim for the grant: *Fusing stem impedance, transfer-function and acoustic signals at multiple heights, with a per-palm learned baseline, detects water stress in arecanut earlier and more reliably than soil moisture alone.* Everything else (localisation of dysfunction, yield gains) follows only if this holds.

3. The product: PhytoHydra

PhytoHydra is one solar-powered node per irrigation block that is strapped to a representative "sentinel" palm, plus a phone dashboard. A farm needs a few nodes, not one per palm: Phytech-style plant sensing typically samples about three plants per 50 to 100 acres (Farm Progress); we plan around one node per 1 to 2 ha block for arecanut, to be confirmed in the field pilot.

What is in the box (v1 field kit)

Part	Qty	What it does
Edge Hub (IP67)	1	Excitation and measurement electronics, edge compute, battery, LoRa radio
Stem Interface Band (N1 to N4)	4	Flexible strap at 4 heights; 2 electrode pods per band (opposite sides of trunk) + stem temperature
Acoustic Sentinel	2	Spring-clamped ultrasonic sensors, low and high on the trunk, for cavitation events
Micro-dendrometer band	1	Invar band + magnetic linear encoder for trunk shrink and swell
Context pack	1	Air T/RH, light, canopy IR temperature, soil moisture probe at 30 cm
Solar panel 6 W + arm	1	Keeps the node running through partial shade
Farm gateway (shared)	1 per farm or FPO cluster	LoRa to 4G bridge; one gateway serves many nodes

What the grower sees (outputs, as in the deck)

Output	Example	Update rate
Hydraulic State Index	0 to 100; "72%: moderate risk"	Every 10 min

Output	Example	Update rate
Where	"Dysfunction 1.2 to 1.8 m above base, north side"	Hourly
Confidence	High / medium / low, from model uncertainty	With every estimate
Trend and forecast	Next 24 to 48 h trajectory, best/likely/worst	Every 3 h
Recovery check	"Recovered 85% of baseline 6 h after irrigation"	After each intervention
Recommended action	"Irrigate Block B, 40 min drip, after 5 pm"	Daily, plus alerts

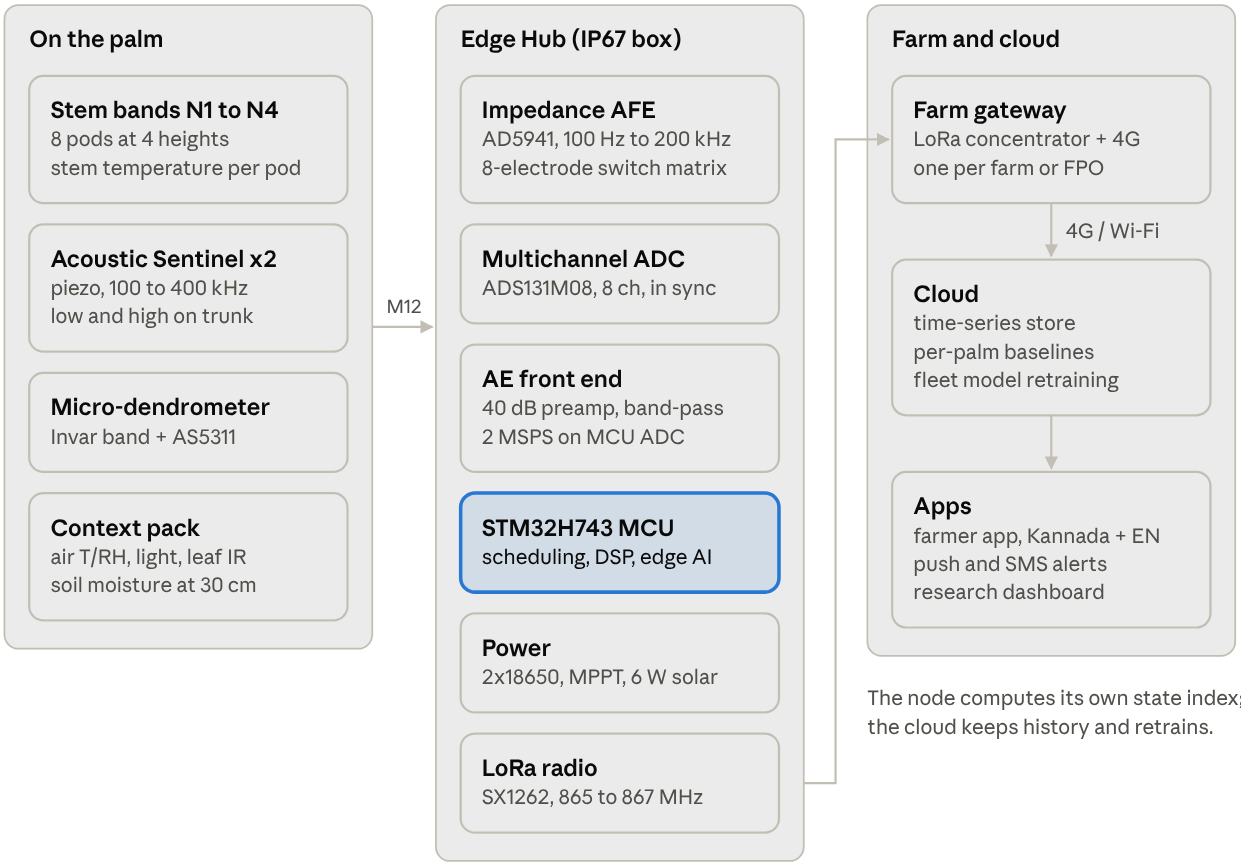
Alerts go out as Kannada and English push messages and SMS, because many plantation owners manage gardens remotely. A web dashboard serves researchers and FPO agronomists with raw signals, per-palm baselines and data export.

What it is not. It is not an image classifier and not a disease diagnosis tool. It reports the palm's hydraulic state and how it is changing; deciding *why* (drought, waterlogged roots, root disease) still needs an agronomist. That honesty matters for credibility with evaluators and farmers.

4. Prototype hardware

The prototype is built in two steps: a **lab rig** from evaluation boards (months 1 to 3) to prove the measurements on seedlings, then a **field node v1** on one custom PCB (from month 4) that survives a plantation for six months. Both use the same sensing chips, so firmware and data carry over.

Each palm node measures, decides at the edge, and reports over LoRa



PhytoHydra v1 system architecture · 4 sensing parts, 6 hub blocks, 3 farm/cloud tiers

Sensors on the trunk run over shielded cables with M12 connectors into the hub; the hub talks LoRa to a farm gateway, which forwards over 4G.

4.1 Subsystem choices

Subsystem	Prototype choice	Key spec	Why this choice
Wideband impedance	Analog Devices AD5941	100 Hz to 200 kHz, 4-wire, low µA excitation	Single-chip impedance spectroscopy, battery friendly
Transfer function + bioelectric potential	TI ADS131M08	8 channels sampled simultaneously, 24-bit, up to 32 kSPS	True simultaneous capture of all segments for H(f) up to ~8 kHz

Subsystem	Prototype choice	Key spec	Why this choice
Electrode switching	2 x ADG1609 (or ADG1208) mux	Low charge injection	Lets one AFE drive any electrode pair
Acoustic emission	10 mm PZT disc, ~150 to 200 kHz resonance, custom housing	40 dB low-noise preamp, 100 to 400 kHz band-pass, comparator wake-up	Cavitation energy sits in the 100 to 300 kHz band (Tyree & Sperry 1989)
Dendrometer	Invar band + ams AS5311 magnetic linear encoder	Sub-micrometre resolution	Palms shrink very little; needs fine resolution
Canopy temperature	MLX90614 IR thermometer aimed at a frond	Non-contact	Leaf-air temperature gap is a classic water-stress index
Air and light	SHT45 (T/RH), VEML7700 (light, PAR proxy)	±0.1 °C, ±1% RH	VPD calculation
Soil moisture	RS-485 capacitive probe at 30 cm	Calibrated to local soil	The baseline every competitor uses; our comparator
MCU	STM32H743 (480 MHz, 2 x 16-bit ADC to 3.6 MSPS)	DSP + TinyML	Samples AE waveforms directly; runs the edge model
Radio	Semtech SX1262 LoRa module	865 to 867 MHz India band	km range through palm gardens, low power
Power	2 x 18650 Li-ion (6.8 Ah), MPPT charger, 6 W panel	~28 days autonomy	Survives monsoon weeks with little sun
Enclosure	IP67 polycarbonate, M12 connectors, stainless band clamps	-10 to 60 °C	Swappable bands; no open cable glands

Lab rig variant: the same AD5941 and ADS131M08 on evaluation boards, a Raspberry Pi 5 as host, and a bench reference AE sensor. Faster to change, and it records raw waveforms that the field node will later summarise.

4.2 The plant interface: the make-or-break part

Arecanut's outer trunk is hard and fibrous, with vascular bundles scattered through it. Contact quality will dominate data quality, so Phase 0 compares two electrode types head to head:

- **Option A, micro-pin electrodes (field default).** Titanium or SS316 pins, 1 mm diameter, inserted 5 to 8 mm into the outer trunk. Stable for months, tiny wound, no drilled probe. Standard in plant electrophysiology and stem resistance work.
- **Option B, Ag/AgCl hydrogel pads (seedlings, short trials).** Fully non-invasive after light surface sanding. Lower damage, but dries out and drifts over days.

Each pod also carries a stem temperature sensor (resistivity drifts about 2% per °C and must be corrected) and an ID chip. Every 10 minutes the node runs a **contact check** at 10 kHz and flags any pod whose contact impedance jumps, so bad data never reaches the model as "stress".

4.3 Measurement cycle (every 10 minutes)

1. Contact check on all 8 electrodes (0.5 s).
2. Impedance spectroscopy: 20 log-spaced frequencies, 100 Hz to 200 kHz, on each of 6 stem segments (3 heights x 2 sides) (about 15 s).
3. Transfer function: inject a multisine (8 tones, 50 Hz to 5 kHz) at N1, record N2 to N4 simultaneously; repeat injecting at N4 (2 s).
4. Passive bioelectric potential: logged continuously at 10 samples/s between pods.
5. Context sensors and dendrometer read (2 s).
6. Acoustic sentinels are always armed; each hit above threshold wakes the MCU, which stores features (amplitude, energy, counts, rise time, peak frequency, arrival-time difference between the two sensors).

Safety for the plant: excitation stays under 10 µA RMS and 1 V, always AC with DC-blocking capacitors, so there is no electrolysis or heating. The deck mentions kHz to MHz; v1 deliberately stops at 200 kHz because long cables swamp signals above ~1 MHz. A MHz front end becomes a v2 item only if Phase 1 shows the extra band adds information.

4.4 Power budget

Load	Duty	Energy per day
Sleep (MCU, regulators)	0.5 mA continuous	12 mAh
Impedance + transfer-function sweeps	144 x 20 s at 120 mA	96 mAh
AE analog wake-up front ends (2)	5 mA continuous	120 mAh
AE hit capture	~2,000 hits x 50 ms at 150 mA	4 mAh
Context sensors + soil probe	144 x 2 s	5 mAh
LoRa uplinks	144 x 0.3 s at 120 mA	1.5 mAh
Total		**~240 mAh/day, about 0.9 Wh**

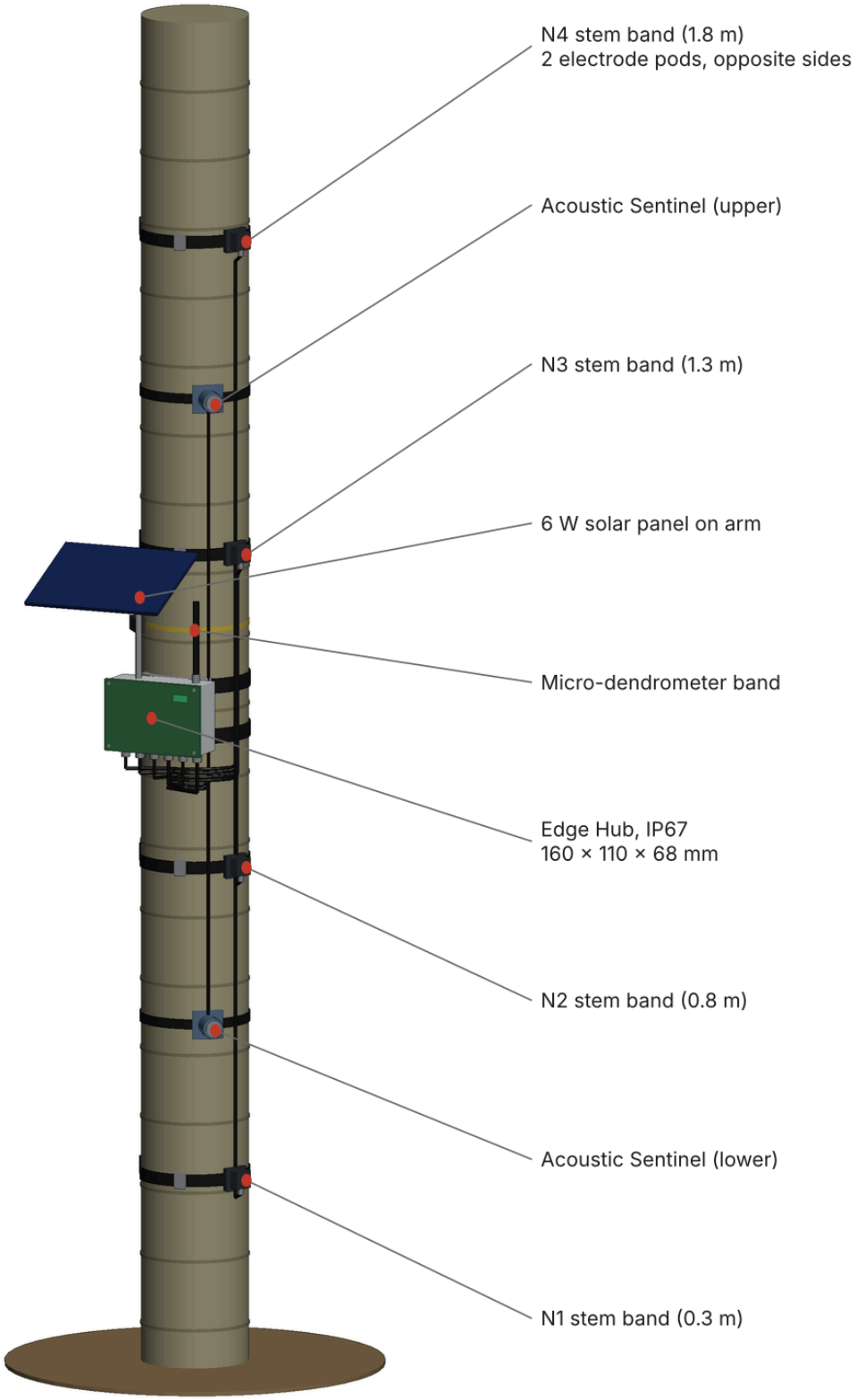
A 6.8 Ah pack gives about 28 days without sun. A 6 W panel yielding even one effective sun-hour under canopy shade (~3 Wh/day after losses) covers the load three times over. These are design estimates; the field pilot measures the real figures.

5. CAD and mechanical design

The mechanical design is a full rev A package: a 3D assembly on a 170 mm arecanut trunk, exploded part views, a dimensioned drawing sheet, parametric OpenSCAD sources for every printed part, and STL meshes. Every trunk-contact surface follows one parameter (`trunk_d`), so the same files fit any palm after a tape-measure reading. It is design intent: nothing has been printed or field-tested yet.

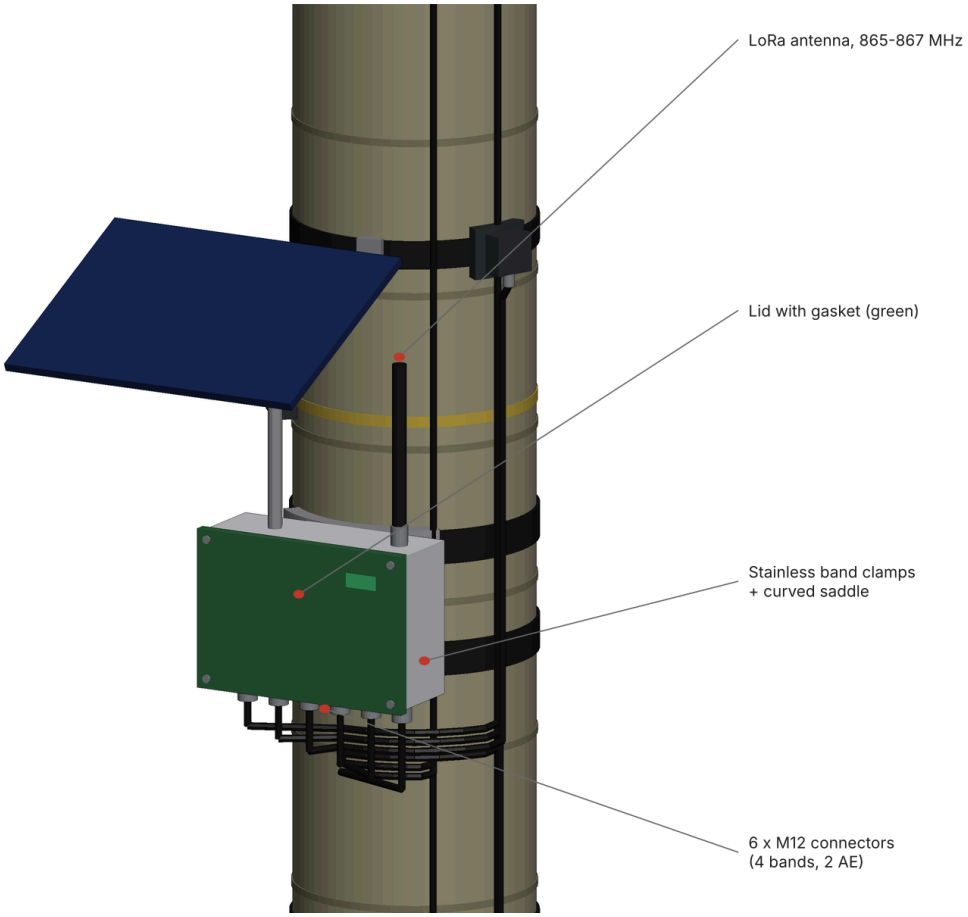
PhytoHydra v1 installed on an arecanut palm

Trunk dia 170 mm (typical mature palm). Band heights for a mature palm; seedling rig uses 5/15/25/35 cm.

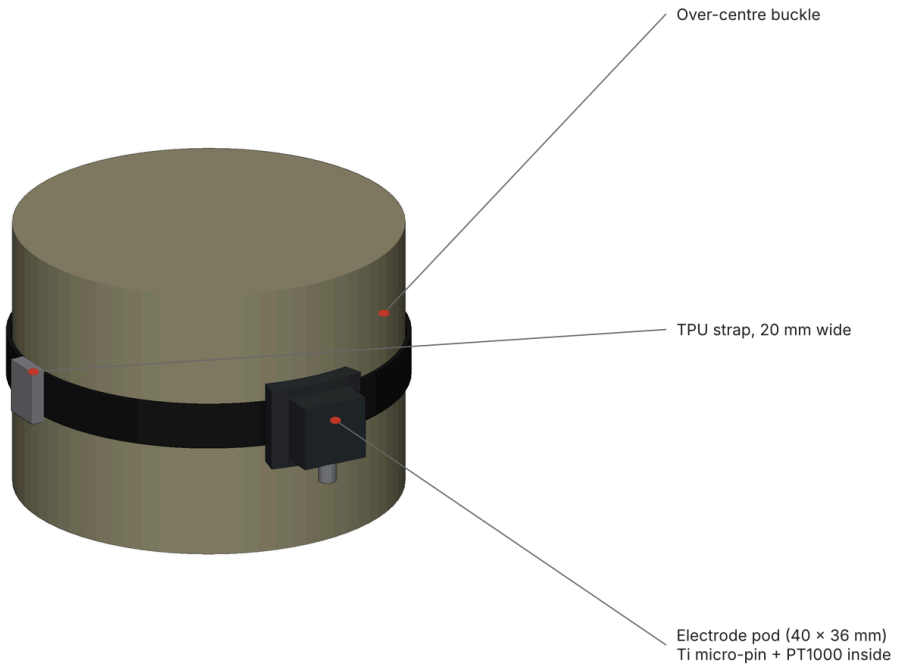


The four bands span 0.3 to 1.8 m, matching the deck's "dysfunction 1.2 to 1.8 m above base" output. The two acoustic sentinels sit on one vertical line so the difference in arrival time of a click tells which half of the trunk it came from.

Edge Hub, solar arm and cable loom (close-up)

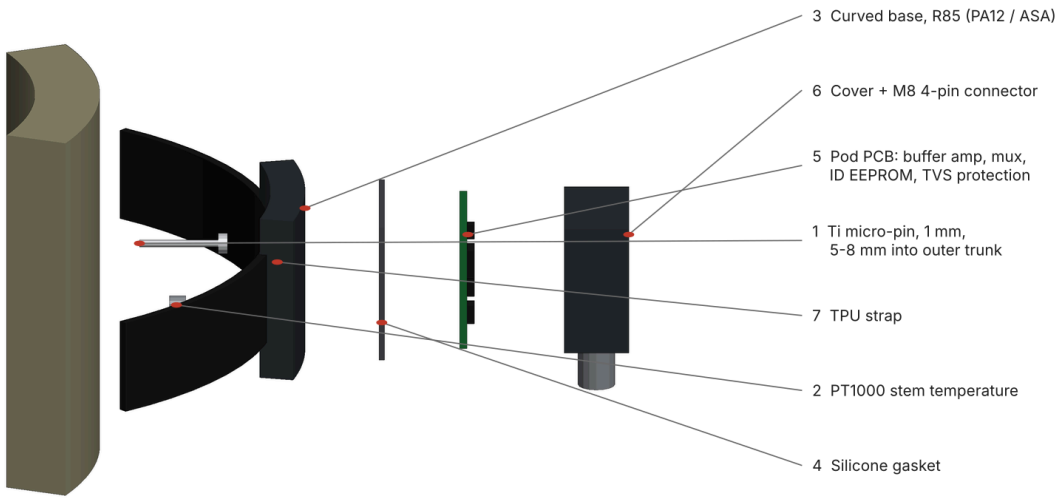


Stem Interface Band: 2 electrode pods on a TPU strap



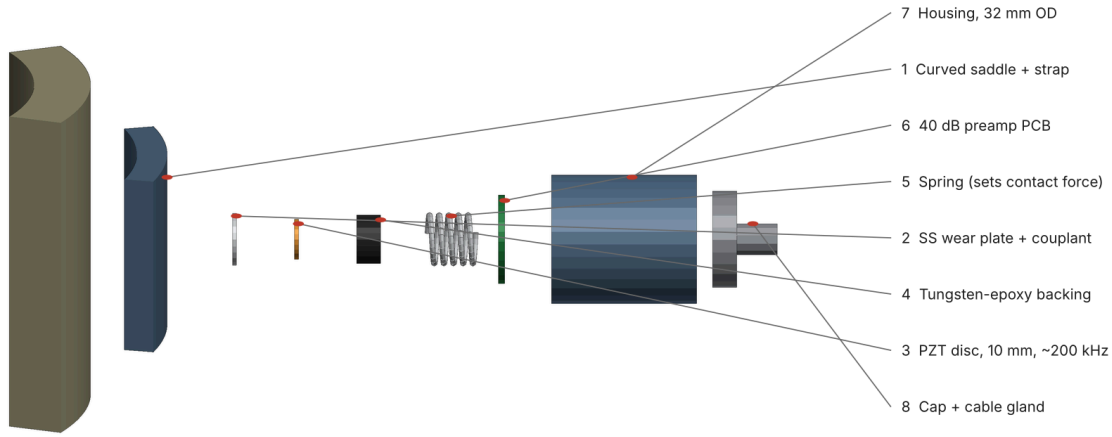
Two pods per band give a west and an east reading at each height, so the model can say which side of the trunk is losing conduction, not just which height.

Electrode pod, exploded (radial axis left to right)

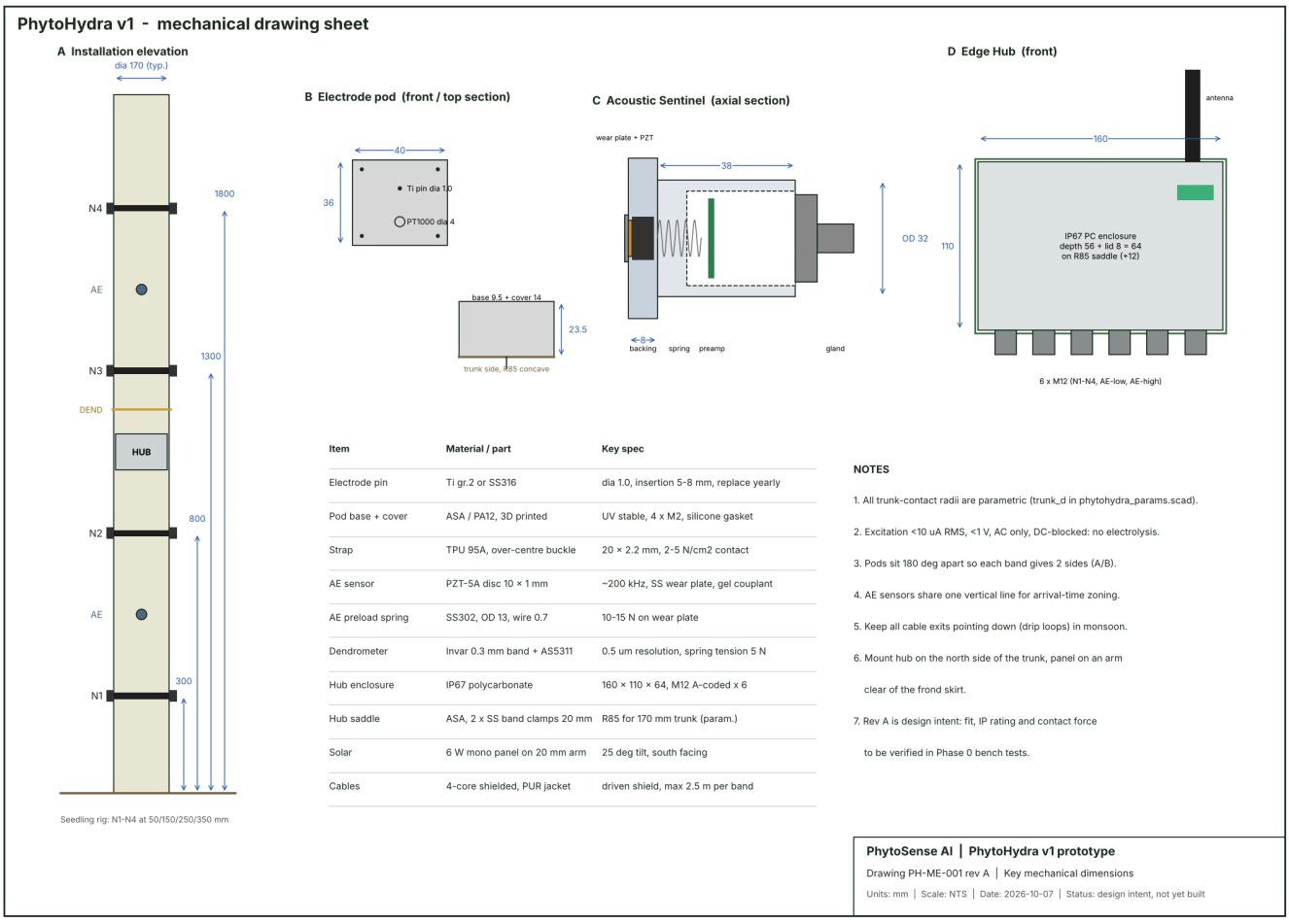


The pod PCB carries a buffer amplifier, an ID EEPROM and TVS protection right at the electrode, so long cables do not pick up noise before the signal is buffered.

Acoustic Sentinel, exploded (radial axis left to right)



The spring sets a repeatable 10 to 15 N contact force. Poor sensor coupling is the most common cause of missed acoustic events (Prometheus protocols).



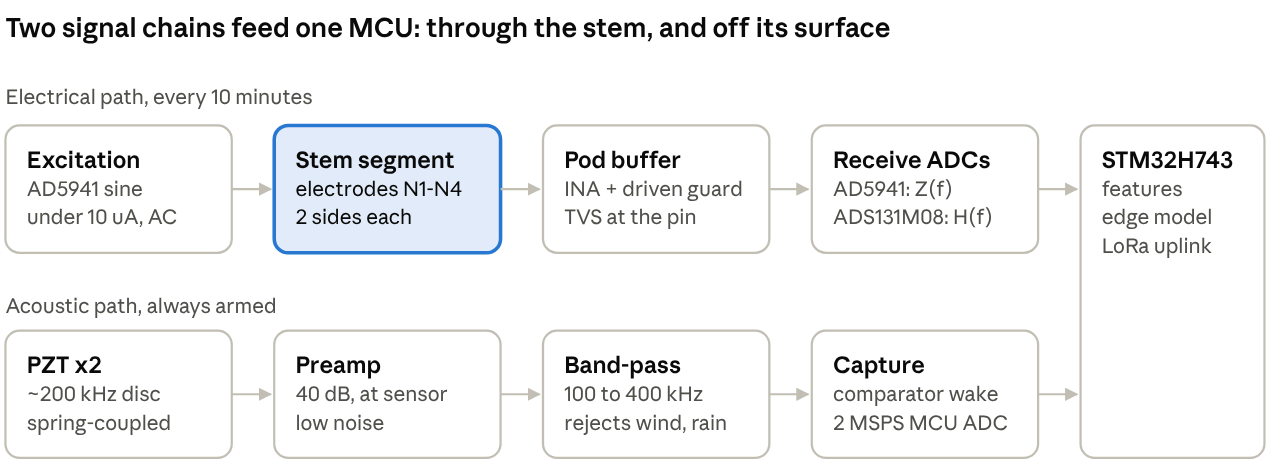
Files in the CAD package (sent alongside this doc as PhytoHydra_v1_CAD_package.zip)

File	What it is	Use
scad/phytohydra_params.scad	Shared parameters: trunk diameter, wall, strap size	Edit once per palm
scad/electrode_pod_base.scad, electrode_pod_cover.scad	Curved pod base with pin hole, PT1000 pocket, PCB pocket, strap slots; cover with M8 boss	3D print in ASA or PA12
scad/acoustic_sentinel_housing.scad, acoustic_sentinel_cap.scad	Saddle + housing with sensor-stack bore, spring cavity; cap with gland	3D print in ASA
scad/hub_saddle.scad	Curved saddle for a 160 x 110 mm IP67 box, M4 insert bosses, band-clamp slots, cable comb	3D print in ASA
stl/*.stl	Assembly on trunk, band, pod, sentinel, hub, exploded views	View, fit-check, slicing reference
render/00_drawing_sheet_PH-ME-001.pdf	Dimensioned drawing sheet	For fabricators and grant reviewers
generator/*.py	Python that builds the meshes and renders	Regenerate after changes

Next mechanical steps: open the .scad files in OpenSCAD (free) and export STL, or import them into FreeCAD and export STEP for a mechanical engineer; print two pods and one sentinel; then measure contact impedance and acoustic coupling on a cut arecanut trunk section in Phase 0.

6. Electronics in detail

The hub runs two independent signal chains that only meet in the MCU, so acoustic hits are never lost while an impedance sweep is running.



PhytoHydra v1 signal chains · electrical (4 stages) and acoustic (4 stages) into one MCU

The palm itself is part of the circuit in the electrical chain: excitation goes in at one electrode, and the stem segment between electrodes shapes what the receive ADCs see.

6.1 Electrical chain

- **4-wire (tetrapolar) measurement.** Current is injected through one electrode pair and voltage is read on a different pair, so electrode contact impedance drops out of the result. This is the single most important choice for stable field data.
- **Two receivers.** The AD5941 does precise impedance spectroscopy one segment at a time. The ADS131M08 samples all 8 electrodes at the same instant, which is what a true transfer function H(f) between heights needs.
- **Pod buffers.** A high-impedance instrumentation amplifier sits inside each pod, with a driven cable shield (guard). Buffering at the electrode stops 2 m of cable from adding capacitance and mains hum.
- **Calibration.** A built-in precision resistor-capacitor network is switched in at every sweep, so drift in the electronics is measured and removed. Stem temperature from each pod's PT1000 corrects resistivity (about 2% per °C).

6.2 Acoustic chain

- Low-noise preamp right at the sensor (40 dB) so the cable carries a strong signal.
- Band-pass 100 to 400 kHz removes wind sway, rain drops and insect noise, which sit far below the cavitation band.
- A comparator with an adjustable threshold wakes the MCU only on real hits, keeping average power low.

- The MCU's own ADC captures each hit at 2 MSPS and keeps features, not raw waveforms: peak amplitude, energy, counts, rise time, peak frequency, and the arrival-time difference between the upper and lower sensors.
- **Calibration** uses the standard pencil-lead break (Hsu-Nielsen source) on the trunk at installation, and a bench comparison against one commercial reference AE sensor.

6.3 Protection and robustness

- TVS diodes on every electrode and sensor line (lightning-induced surges are common in monsoon storms).
- Galvanic isolation between the measurement ground and the solar/battery side, so charging current never flows through the palm.
- Conformal-coated PCBs, desiccant pack and a breather vent in the IP67 box.
- Watchdog, brown-out reset and over-the-air firmware updates through the gateway.

6.4 PCB plan

- **Main board** (4-layer, about 100 x 80 mm): analog island (AD5941, ADS131M08, muxes, reference) separated from the digital and radio island, star ground, shielded LoRa section.
- **Pod board** (2-layer, 30 x 26 mm): buffer amplifier, PT1000 interface, ID EEPROM, TVS.
- **AE preamp board** (2-layer, 22 mm round): preamp and first filter stage.
- **Bring-up order:** lab rig on evaluation boards (month 1) → main board spin 1 (month 3) → spin 2 with field fixes (month 5).

7. Firmware, data and the AI engine

The AI is a dynamic state estimator, not an image classifier: it learns each palm's own healthy rhythm, then tracks hidden hydraulic states and how they move. It is built in four stages so that something useful works on day one, before any labelled data exists.

7.1 Pipeline (as in deck slide 6, made concrete)

1. **Quality gate.** Drop any sweep where contact impedance jumped, the calibration network drifted, or rain was detected on the trunk. Bad data is the main risk in field electrophysiology.
2. **Signal processing.** Notch filter 50 Hz mains, detrend, temperature-correct impedance to 25 °C, fit a Cole model to each impedance spectrum.

3. **Features per segment, every 10 minutes** (table below).
4. **Per-palm baseline.** For each feature, segment and time of day, keep a slowly adapting mean and spread learned over the first 7 to 14 well-watered days. Deviation is a z-score.
5. **Multimodal fusion.** Combine electrical, acoustic, dendrometer and context deviations into one evidence vector.
6. **State estimation.** A state-space model tracks hidden states with uncertainty.
7. **Forecast and recommendation.** Simulate the next 24 to 48 h under "irrigate now", "irrigate tonight" and "do nothing", and pick the cheapest action that keeps the stress index under the farmer's threshold.

```
z_{k}(t) = \frac{x_{k}(t) - \mu_{k}(\text{hour}, \text{palm})}{\sigma_{k}(\text{hour}, \text{palm})}
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x_{t+1} = F(x_t, u_t, e_t) + w_t, \qquad y_t = G(x_t) + v_t
```

Here x is the hidden state (stem hydration W, hydraulic conductance K, embolism load E, recovery capacity R), u is irrigation, e is weather (VPD, light, temperature), y is what the sensors measure, and w, v are noise.

7.2 Features

Group	Features	Physiological meaning
Impedance spectrum	Cole parameters R0, R∞, α, fc; R0/R∞ ratio; phase at 10 kHz	Water outside vs inside cells; membrane state
Transfer function	Gain and phase of H(f) between heights, coherence, group delay	How conduction changes along the stem
Asymmetry	Side A vs side B difference per height	One-sided root or vascular problem
Bioelectric potential	Daily amplitude, variance, step responses after irrigation	Plant electrical response to stress
Acoustic	Hits per hour, energy, 100 to 200 kHz cluster share, upper vs lower ratio	Cavitation events, where they start

Group	Features	Physiological meaning
Dendrometer	Daily shrink amplitude, overnight recovery, trend	Stem water storage
Thermal	Canopy minus air temperature, normalised by VPD	Stomatal closure
Context	VPD, light, soil moisture, time since irrigation	Demand and supply

7.3 Four model stages

Stage	When	Model	Needs labels?
1. Baseline deviation	Day 1	Per-palm z-scores + rule fusion, runs on the MCU	No
2. Supervised stress estimate	After Phase 1 (month 6)	Gradient-boosted trees on features, trained against leaf water potential and pot-weight transpiration	Yes, from Phase 1
3. State-space tracking	Month 7 to 9	Extended Kalman filter around a simple stem hydraulic model (resistance + water storage), parameters learned per palm	Partly
4. Forecast + counterfactual	Month 9 to 12	The same model run forward under different irrigation plans	Uses stage 3

The hydraulic model is deliberately small (a few parameters per palm) because 24 seedlings and 10 field palms are a small dataset. A large deep network would overfit; physics plus a light learned layer will not.

7.4 Edge vs cloud

- **On the node (STM32H743):** quality gate, features, baseline z-scores, stage 1 index, alerts. Works with no connectivity.

- **In the cloud:** stage 2 to 4 models, per-palm parameter fitting, fleet-wide retraining, dashboards. Updated models are sent back to nodes over LoRa as small parameter files.

7.5 Data scale

Phase 1 alone produces about 1.9 million impedance spectra (18 instrumented plants x 6 stem segments x 144 sweeps a day x about 120 days), plus acoustic hit records and continuous context data, all time-aligned with manual ground truth. This labelled arecanut dataset is a core asset: nobody else has it.

8. Bill of materials and unit cost

One complete prototype node costs about **₹33,500** in parts at a 10-unit build, falling to about **₹15,300** at 1,000 units. Prices are estimates from typical Indian distributor and online pricing in 2026, not quotes; get two quotes for every line above ₹1,000 before the grant budget is final.

Item	Qty per node	Prototype (10 units), ₹	At 1,000 units, ₹
Main MCU STM32H743 (+ crystal, flash)	1	1,100	650
Impedance AFE AD5941 + reference + calibration network	1	1,200	600
Sync ADC ADS131M08	1	900	520
Analog switches ADG1609 x2	1 set	800	450
Main PCB 4-layer + assembly	1	2,500	700
Electrode pods (PCB, INA, PT1000, EEPROM, TVS, M8, print)	8	6,000	2,640
Ti micro-pin electrodes (incl. spares)	10	800	400
TPU straps + buckles	4	1,000	360
Acoustic sentinel (PZT, backing, spring, preamp PCB, housing)	2	3,600	1,400

Item	Qty per node	Prototype (10 units), ₹	At 1,000 units, ₹
Micro-dendrometer (Invar band + AS5311 + magnet strip)	1	1,900	1,000
Context sensors (SHT45, VEML7700, MLX90614)	1 set	1,700	850
Soil moisture probe RS-485, 30 cm	1	3,200	1,600
LoRa SX1262 module + antenna	1	1,000	450
Battery 2 x 18650 + BMS + MPPT charger	1	1,400	800
Solar panel 6 W + arm	1	1,100	650
IP67 enclosure + saddle + clamps	1	1,500	700
M12 connectors (6 pairs) + shielded PUR cable	1 set	3,300	1,300
Misc (vent, desiccant, conformal coat, fasteners)	1	500	250
Total per node		**33,500**	**15,320**

Shared and one-off items (not per node)

Item	Cost, ₹	Note
Farm gateway (LoRa concentrator + 4G + solar)	~25,000	One per farm or FPO cluster
Lab rig evaluation boards (AD5941, ADS131M08, Raspberry Pi 5)	~45,000	One-time, Phase 0 to 1
Commercial reference AE sensor + preamp	~70,000	Calibrates the low-cost sentinels; borrow if a partner lab has one
Pot load cells for gravimetric ground truth (HX711 + 50 kg cell)	~600 each	24 needed for Phase 1

Cost-down path for the commercial "Farm node" (v2): drop the dendrometer from standard farm nodes (keep it in a research kit), merge pod electronics onto a flex PCB, and buy connectors in volume. Target parts cost about **₹12,000 to 14,000** at 1,000 units, which supports the pricing in section 13.

9. Proof of concept and validation protocol

The PoC must answer one question with numbers: **how many hours earlier than soil moisture, and earlier than visible symptoms, does PhytoHydra reliably detect water stress in arecanut, and how often does it raise false alarms?** Three phases get there.

Phase 0: bench (months 1 to 2)

- Electrode comparison on cut arecanut trunk sections and potted seedlings: Ti pin vs SS316 pin vs Ag/AgCl gel. Measure contact impedance, drift over 14 days and noise.
- Acoustic sensor coupling: pencil-lead breaks at 10 to 100 cm from the sensor to measure attenuation along the palm trunk, low-cost sentinel vs reference sensor.
- Dehydrating cut segments on a balance: impedance and AE vs measured water loss. This gives the first calibration curve.
- **Gate:** contact impedance stable within ±10% over 14 days; low-cost AE sensor catches at least 80% of the hits the reference sensor catches.

Phase 1: controlled stress on seedlings (months 2 to 6)

- **Plants:** 24 two-year-old arecanut seedlings (deck range 20 to 30), same variety and nursery, in 50 L pots under a rain shelter.
- **Design:** randomised blocks; 18 instrumented, 6 un-instrumented to check the electrodes themselves do not harm growth.
- **Treatments, each run as a drying and re-watering cycle, repeated 3 times:**

Group	n	Irrigation	Purpose
Control	6	100% of daily pot water loss	Healthy baseline, false-alarm rate
Mild deficit	4	60% of loss	Sensitivity at small stress
Moderate deficit	4	30% of loss	Main detection test

Group	n	Irrigation	Purpose
Severe, then re-water	4	Water withheld to visible wilting, then 100%	Lead time to symptoms, recovery tracking
Split-root (one side dry)	6 (incl. 2 un-instrumented)	One half of a divided pot dried	Can the A/B pods localise a one-sided problem?

- **Ground truth (independent of our sensors):**
  - Pot weight on load cells every 10 min: actual transpiration, the most objective daily stress signal.
  - Midday leaf water potential with a pressure chamber, 3 times a week (borrowed or partner lab).
  - Stomatal conductance with a porometer, 3 times a week.
  - Soil moisture at two depths, canopy temperature, daily photos scored blind for visible symptoms.
- **Stress onset definition (fixed before the experiment):** the first day a plant's transpiration falls below 80% of the control mean, or its leaf water potential falls 0.3 MPa below the control mean, whichever comes first.

Phase 2: field pilot (months 5 to 12)

- 10 nodes in one irrigated arecanut garden in Shivamogga or Dakshina Kannada, across two irrigation blocks, installed by February so the March to May dry season is captured.
- 2 of 10 nodes on palms with known trunk or root problems (e.g. stem bleeding or basal stem rot noted by the farmer) vs healthy neighbours, as a case-control test of localisation on mature trunks.
- Weekly leaf water potential on 6 palms with the partner institute; farmer's irrigation log; soil moisture profile.
- In block A, the farmer follows PhytoHydra alerts for 6 weeks; block B stays on the normal schedule. Compare water applied and palm stress.

Metrics and success criteria

Every model is evaluated on plants and cycles it was not trained on (leave-plants-out cross-validation), so the numbers reflect a new palm, not memorisation.

Metric	Definition	Success threshold (to fix in the protocol)
Lead time vs soil moisture	Hours between PhytoHydra alert and soil-moisture threshold alert	Median at least 12 h earlier
Lead time vs visible symptoms	Hours between alert and first blind-scored symptom	Median at least 24 h earlier
Detection rate	Stressed plant-cycles detected before onset + 24 h	At least 80%
False alarms	Alerts on control plants	At most 1 per plant per 2 weeks
Localisation	Correct side in split-root plants	At least 75%
Ablation	Approaches A to D in deck slide 7: soil+weather, single-point electrical, multi-point electrical, multi-point + acoustic	D beats A significantly (paired test)
Plant safety	Growth of instrumented vs un-instrumented plants	No significant difference

The deck's table promises lead times of 24 to 72 h for approach D. Keep those as hypotheses to test, not results; the thresholds above are what a reviewer can hold the team to.

What the grant period will not prove: yield impact. Arecanut yield needs at least one full season and many palms. State this openly and propose it as the GENESIS Pilot-stage objective.

10. Market

Arecanut is a big, concentrated, high-value first market: India grew about 14 lakh tonnes on 9.49 lakh ha in 2023-24, and Karnataka alone had 6.76 lakh ha and 10 lakh tonnes (Outlook Business). India is about 63% of world production (same source).

Correction for the deck: slide 8 says "~2.9 lakh ha under arecanut in Karnataka". The current figure is about 6.76 lakh ha (2023-24); Karnataka's own estimate for 2025-26 is about 7.56 lakh ha (IJIRT 2025). Fix this before the interview; evaluators check.

10.1 Why growers can pay

- India's average yield was about 1,470 kg/ha in 2023-24 (IJIRT 2025).
- Shivamogga APMC modal prices ran about ₹39,000 to ₹54,000 per quintal in mid-2026 (acrop.app).
- At ₹400/kg, an average hectare grosses about **₹5.9 lakh a year**. The PhytoHydra cost per hectare (section 13) is under 2% of that, so protecting about 2% of yield, or saving pumping and labour, pays for it.

10.2 Market sizing (arecanut only, one node per hectare)

Level	Definition	Hectares (nodes)	Subscription value / yr	Hardware value (one-time)
TAM	All arecanut in India	9.49 lakh	₹569 Cr	₹2,088 Cr
Karnataka	All arecanut in Karnataka	6.76 lakh	₹406 Cr	₹1,487 Cr
SAM	Irrigated, commercially run Karnataka gardens of 2 ha or more (assumed 25% of area)	1.69 lakh	₹101 Cr	₹372 Cr
SOM, year 5	2,000 farm nodes + 10 research deployments	2,000	₹1.2 Cr ARR + research ₹0.3 Cr	₹4.4 Cr cumulative

Assumptions: ₹6,000 per node per year subscription and ₹22,000 hardware price (section 13); one node per hectare; the 25% SAM share is an assumption to replace with survey data from FPOs in Phase 2.

10.3 Wider context

- India's precision farming market is estimated at roughly US$206 to 424 million in 2025, depending on the analyst, growing 9 to 16% a year (MarketsandMarkets, IMARC). The spread itself says these are rough; use them only as context.
- Expansion crops in the deck's roadmap (coconut, coffee, banana, then mango, citrus, grapes) multiply the addressable area several times, using the same hardware with crop-specific baselines.

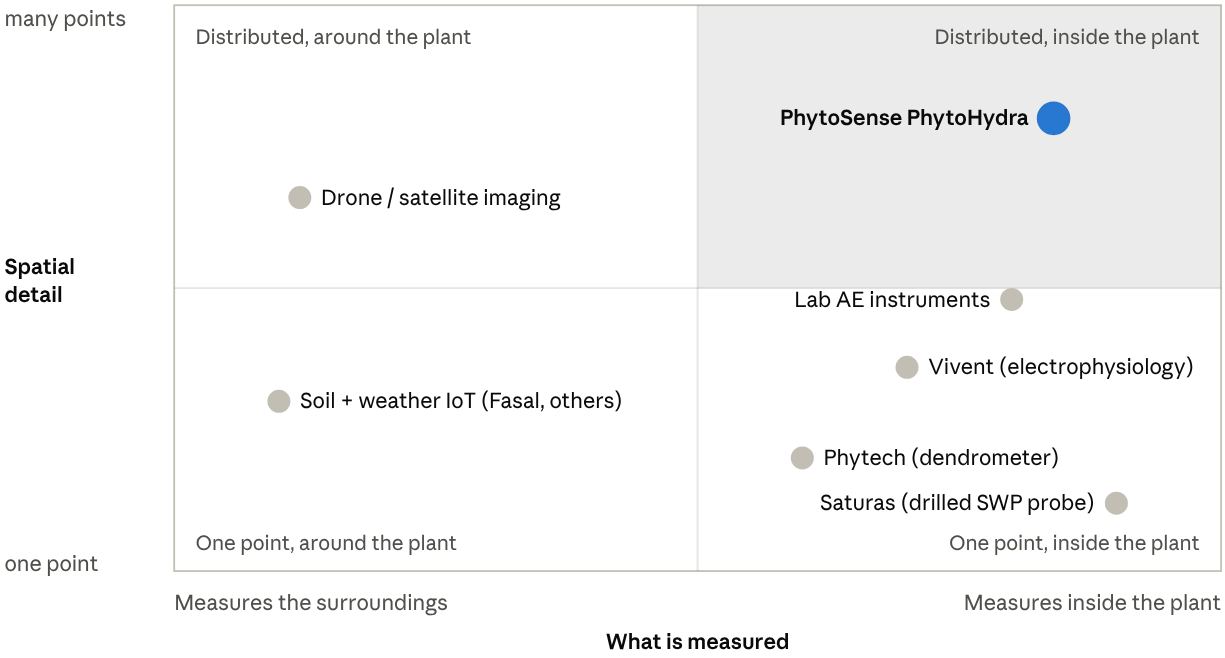
10.4 Customer segments

Segment	Who	What they buy	When
Research institutes	ICAR-CPCRI (Vittal, Kasaragod), UAHS Shivamogga, Directorate of Arecanut and Spices Development, agri universities	Research kits, data access, co-authored trials	Year 1 (first revenue)
Progressive growers	2 to 20 ha irrigated gardens, often absentee owners	Farm nodes + subscription	Year 2
FPOs and co-operatives	Arecanut co-operatives and FPOs in Shivamogga, Dakshina Kannada, Uttara Kannada	Cluster deployments, shared gateways, advisory dashboards	Year 2 to 3
Estates and nurseries	Large estates; nurseries and breeding programmes screening drought tolerance	Many nodes; phenotyping platform	Year 3
Partners	Drip irrigation companies, agri-input firms, crop insurers	API and data licensing, co-branded kits	Year 3 onward

11. Competitors and positioning

Plant-based sensing is a proven, funded category, which de-risks the idea; no one offers distributed multi-signal stem sensing for palms at Indian prices, which is the opening.

Nobody else measures inside the palm at several points along the stem



Competitive positioning · qualitative placement from public product descriptions

Placements are qualitative, from each company's public description; they show where the empty quadrant is, not a measured ranking.

Company	Approach	Proof it is a real market	Gap PhytoHydra fills
Vivent (Switzerland)	Plant electrophysiology biosensors, AI diagnosis	Raised €7.5 million in Nov 2025; world's largest plant electrophysiology library (per the company)	Mainly greenhouse crops in Europe and North America; single point, no hydraulic localisation
Phytech (Israel)	Dendrometers + soil + weather, irrigation as a subscription service	Long-running per-acre subscriptions; historically about US$20 per acre (Farm Progress)	Palms barely change diameter; no electrical or acoustic layer

Company	Approach	Proof it is a real market	Gap PhytoHydra fills
Saturas (Israel)	Miniature stem water potential sensor embedded by drilling	EU-funded trials in Israel, Spain, California	Drilled, one point, imported cost (early estimate about US$300 per sensor)
SupPlant (Israel)	Plant and soil sensors with AI irrigation advice	Named as Saturas's competitor (PitchBook)	Not palm-specific
Fasal (India)	Soil and weather IoT + vernacular advisory SaaS for horticulture	Farmers buy IoT upfront, then pay SaaS; reported 2024 revenue about US$4.1 million (Preqin)	Measures around the plant, not inside it. Also a likely **partner**: PhytoHydra could feed plant signals into existing advisory platforms
Lab AE instruments (Physical Acoustics, Vallen)	Ultrasonic acoustic emission for research	Standard in plant hydraulics labs since Tyree & Sperry (1989)	Research-only, expensive, no field product

Defensible advantages to build: (1) the palm electrode interface and contact-quality system; (2) the multi-height transfer-function method fused with acoustic validation, which is the patentable core; (3) the first labelled arecanut hydraulic dataset; (4) Kannada-first farmer UX and FPO distribution, which foreign players lack.

12. Real-world use cases

The first paying use is dry-season irrigation scheduling; the others reuse the same node and data, and each is a hypothesis until the pilot shows it.

#	Use case	Who	What PhytoHydra does	Status
1	Dry-season irrigation scheduling	Irrigated gardens, Feb to May	Tells when and how long to run drip per block from palm stress, instead of a fixed 16 to 20 L/palm/day or 175 L every 4 to 7 days (DASD)	Core; validated in Phase 1 and 2
2	Borewell and pump planning	Plateau and dry-district gardens	Shows which blocks suffer first when water runs short, so scarce water goes where it protects most yield	Phase 2
3	Recovery check after irrigation or fertigation	All users	"Did it work?" recovery curve 6 to 24 h after an intervention	Phase 1 (re-watering cycles)
4	Monsoon waterlogging alert	Low-lying gardens	Waterlogged roots also cut water uptake; a falling conduction signal with wet soil points to drainage, which extension advice already stresses (KAU)	Hypothesis, monsoon data from Phase 2
5	Drought-tolerance phenotyping	Breeders, CPCRI, nurseries	Ranks seedlings or varieties by how fast they lose and recover hydraulic function	Phase 1 data is itself a pilot of this

#	Use case	Who	What PhytoHydra does	Status
6	Research support on palm decline	Plant pathologists	Continuous hydraulic records on palms with yellow leaf disease or root problems vs healthy ones. Yellow leaf disease has destroyed over 16,000 ha in Dakshina Kannada per CAMPCO (Deccan Herald)	Research only; no diagnostic claim
7	Evidence for crop insurance	Insurers, FPOs	Plant-level stress records to support weather-index claims; Shivamogga's MP noted many district rain gauges were not working (Deccan Herald)	Year 3+
8	Closed-loop irrigation	Drip companies	API to switch drip valves automatically	Year 3+

A day with PhytoHydra (illustrative, April, a 4 ha garden near Sagar)

- **06:00** Overnight, the hub logged full recovery in Block A but only partial recovery in Block B: the dendrometer did not return to its baseline and N1 to N2 conduction stayed 1.8 standard deviations low.
- **11:30** Acoustic hits on the lower sentinel in Block B rise to three times the usual rate for that hour. Soil moisture at 30 cm still reads 21%, which a soil-only system calls "fine".
- **12:10** The owner, who lives in Bengaluru, gets a Kannada push message: "Block B palms under moderate water stress (index 64, confidence high). Supply problem near the base. Check drip line B for blockage; if clear, run 45 min after 5 pm."
- **13:00** The farm worker finds two clogged laterals in Block B and flushes them.

- **Next morning** Recovery curve back to 92% of baseline; the app logs "intervention worked" and the agronomist dashboard at the FPO sees the event.

13. Business model and go-to-market

Sell or lease the node, charge a yearly subscription per node, and earn early revenue from research institutes while farm adoption builds. The deck's four revenue streams stay; here they get prices.

13.1 Pricing (targets, to test with growers in Phase 2)

Offer	Price	Includes
Farm node (buy)	₹22,000 one-time	Node, 4 bands, 2 acoustic sentinels, context pack, installation
Monitoring subscription	₹6,000 per node per year	App, alerts, recommendations, Kannada/English support, model updates
Farm node (lease)	₹12,000 per node per year, 3-year term	Everything above, no upfront cost, replacement included
Farm gateway	₹25,000 per farm, or included for FPO clusters	LoRa to 4G bridge for many nodes
Research kit	₹45,000 per node	Adds dendrometer, raw waveform access, data export
Research platform licence	₹1.5 lakh per institution per year	Dashboard, experiment design tools, API
Data and API licensing	Negotiated	Anonymised datasets and APIs for irrigation, input and insurance partners

13.2 Unit economics at 1,000 units a year

Line	Per node
Hardware price	₹22,000

Line	Per node
Hardware cost (₹13,000 parts + ₹1,500 assembly and test + ₹1,000 install)	₹15,500
Hardware gross margin	₹6,500 (about 30%)
Subscription per year	₹6,000
Cost to serve per year (cloud, connectivity, agronomy support)	about ₹1,500
Subscription gross margin	₹4,500 (75%)
5-year gross profit per node	about ₹29,000
Target acquisition cost via FPO and dealer channel	under ₹5,000

For the grower: buying costs about ₹10,400 a year per hectare over five years (₹22,000 / 5 + ₹6,000), about 1.8% of the roughly ₹5.9 lakh gross per hectare in section 10. That is the number to show farmers, with the pilot's measured water and stress results beside it.

13.3 Go-to-market

1. **Year 1, credibility:** research partnership with one institute (ICAR-CPCRI or UAHS Shivamogga) for ground truth and co-authored results; 2 lighthouse growers who host the field pilot.
2. **Year 2, first farm sales:** demonstrations through Krishi Vigyan Kendras and arecanut co-operatives such as CAMPCO and TSS Sirsi; leasing to remove the upfront barrier; one FPO cluster with a shared gateway.
3. **Year 2 to 3, channels:** drip irrigation dealers bundle PhytoHydra with new drip installs; advisory platforms (such as Fasal) as integration partners rather than rivals.
4. **Year 3 onward, expansion:** coconut and coffee (same palms and plantation belts), then banana and fruit trees; insurer and data partnerships once the dataset is large.

Government support to target along the way: GENESIS Pilot (up to ₹40 lakh) and Matching Investment (up to ₹50 lakh) after EIR (IC IIT Patna summary); horticulture mission schemes that already fund arecanut programmes in Karnataka (Krishi Jagran).

14. Grant fund utilisation and 12-month plan

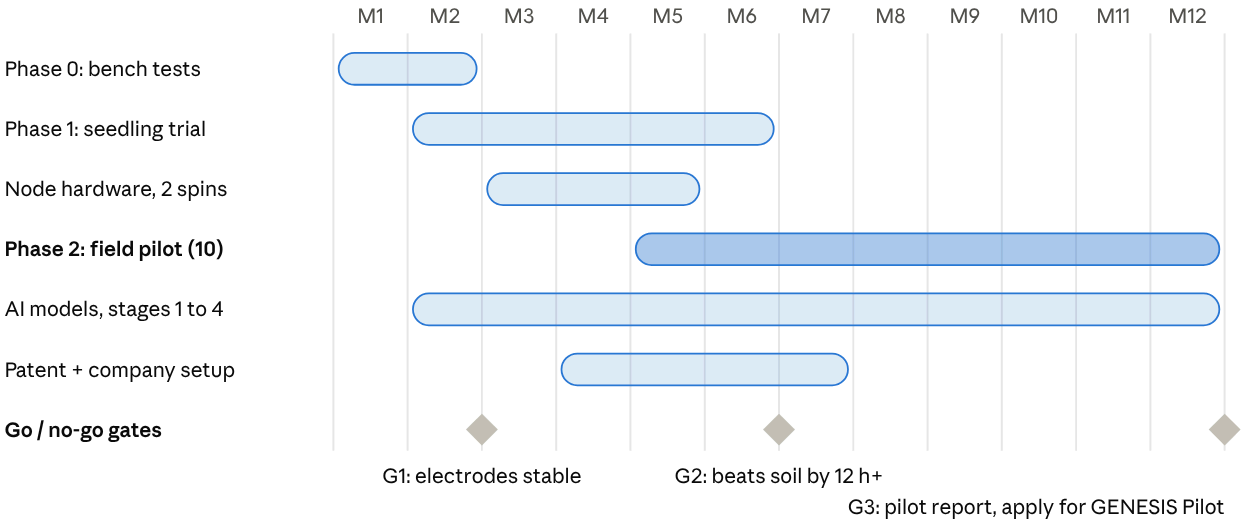
The full ₹10 lakh EIR grant goes to evidence: about one third buys the 10 field nodes, and most of the rest pays for experiments and ground truth. The application left the Fund Utilisation Plan field blank; this table can be pasted into it.

Head	₹	What it buys
Field nodes	3,35,000	10 PhytoHydra v1 nodes at about ₹33,500 each (section 8)
Lab rig and bench tools	70,000	AD5941 and ADS131M08 evaluation boards, Raspberry Pi 5, bench fixtures
Reference acoustic sensor	70,000	One commercial AE sensor + preamp to calibrate the low-cost sentinels (moves to contingency if a partner lends one)
PCB iterations	60,000	2 spins of the main board, pod and preamp boards, stencils
Seedling experiment	70,000	24 seedlings, 50 L pots, media, rain shelter, drip kit, 24 load cells
Ground truth and partner lab	80,000	Pressure chamber and porometer access, partner fees, consumables
Gateway, connectivity, cloud	45,000	1 farm gateway, 12 months 4G and cloud hosting
Compute and software	25,000	Model training credits, tools
Travel, installation, field help	90,000	Kozhikode to Karnataka trips, installation days, local field assistant
IP and company	45,000	Prior-art search, provisional patent, incorporation and DPIIT registration
Contingency (11%)	1,10,000	Breakage, monsoon damage, price changes
Total	**10,00,000**	

Check with the incubator which cost heads the EIR grant allows (some cohorts restrict capital equipment or pay part as a fellowship) and re-balance before submitting.

Twelve months from bench to a field-validated prototype, three gates

Month 1 = grant start; field nodes must be on palms before the March to May dry season



12-month execution plan · 6 workstreams, 3 gates

If G1 fails, switch electrode type before building field nodes. If G2 fails, the field pilot still runs but narrows to the signals that did beat soil moisture.

15. Risks, and gaps to fix in the deck and application

15.1 Fix before the screening interview

#	Gap found	Where	Fix
1	Fund Utilisation Plan, Prototype Images, Demo video, One-pager, Key milestones, IP status all left blank	Application	Paste section 14's table; attach the CAD renders and drawing sheet from section 5; record a 2-minute bench demo once the lab rig runs
2	"~2.9 lakh ha under arecanut in Karnataka"	Deck slide 8	About 6.76 lakh ha (2023-24), see section 10
3	KPIs stated as facts: 20-30% water saved, 10-25% less yield loss, 12-72 h earlier	Deck slides 7 and 9	Label as "targets to validate"; show the success thresholds in section 9

#	Gap found	Where	Fix
4	"Computer Vision" listed as a technology	Application	Either remove it, or say it is only used to score visible symptoms as ground truth
5	Application mentions thermal responses; deck shows no thermal sensor	Both	Add the canopy IR sensor (section 4) to the deck's architecture slide
6	Roadmap starts "2025 Foundation"	Deck slide 10	Shift to 2026-27 (PoC), 2027-28 (pilot), 2028+ (scale)
7	Market figures without sources (₹80,000 Cr plantation market, US$12B global precision ag)	Deck slide 9	Cite a source or replace with the sourced sizing in section 10
8	Not incorporated, not DPIIT-registered; founder city is Chennai	Application	Confirm with the host incubator that a final-year student applicant is eligible and how the Tier-II/III rule applies with physical incubation in Kozhikode; budget incorporation (section 14)
9	The application PDF contains Aadhaar and PAN numbers	Application export	Share a redacted copy whenever the PDF goes anywhere other than the MeitY portal
10	Solo founder, no named advisors	Application	Name a plant physiology advisor and a hardware mentor before the interview (section 16)

15.2 Risk register

Risk	Likelihood	Impact	Mitigation
Palm trunk anatomy gives weak or unstable electrical contrast	Medium	High	Phase 0 electrode tests and dehydration curves first; acoustic and dendrometer as independent signals; gate G1 before building field nodes
Temperature and rain swamp the electrical signal	High	Medium	PT1000 per pod, temperature correction, rain-flag quality gate, per-hour baselines
Acoustic noise (wind, insects, rain) causes false hits	Medium	Medium	100 to 400 kHz band-pass, frequency clustering, two-sensor coincidence
Electrodes harm the palm	Low	High	Tiny pins, AC under 10 µA, un-instrumented control plants in Phase 1
Field damage: rodents chewing cables, monkeys, theft, lightning	Medium	Medium	PUR armoured cable, cable guards, TVS protection, hub mounted high, spares in budget
Monsoon humidity kills electronics	Medium	High	IP67, conformal coat, breather vent, desiccant, bottom cable exits with drip loops
Growers find it too expensive	Medium	High	Leasing, FPO cluster pricing, show measured savings from Phase 2
Arecanut demand risk (price swings, health-policy debate around areca)	Medium	Medium	Platform is crop-agnostic; coconut and coffee next on the same hardware
Too little data for robust AI	High	Medium	Small physics-based state model, per-palm baselines, leave-plants-out testing
Solo-founder bandwidth	High	High	Advisors, one hardware intern, partner institute runs ground-truth measurements

16. Roadmap, IP, team and next steps

16.1 Roadmap beyond the grant

Period	Stage	Goal
Months 1-12 (2026-27)	EIR PoC	TRL 3 to TRL 5: lab-proven and field-tested on 10 palms
2027-28	GENESIS Pilot (up to ₹40 lakh)	100 nodes across 3 districts; first full-season yield and water data; first paying research customers
2028-29	Commercial launch	Farm node v2 (cost-down), arecanut + coconut, FPO channel, leasing
2029-30	Platform	Coffee and banana; API for irrigation and insurance partners
2030+	Scale	Multi-crop, multi-state, dataset and model licensing

16.2 Intellectual property

- **File a provisional patent before any public demo, paper or competition pitch** (Indian provisional gives 12 months to file the complete specification). Candidate claims:
  1. Method of estimating and localising hydraulic dysfunction in a plant stem by fusing multi-height electrical transfer functions with acoustic emission arrival times, against a per-plant learned baseline.
  2. Curved electrode pod for monocot trunks with integrated stem temperature compensation and periodic contact-quality self-diagnosis.
- Keep trained models, feature pipelines and the dataset as trade secrets; publish only aggregated science, after filing.
- Search existing patents first (Saturas, Vivent and acoustic irrigation filings exist), so the claims are drafted around them.

16.3 Team to assemble

Role	Why	How
Founder (Ajith Anand R)	ECE: electronics, firmware, AI	Full-time from grant start, per application
Plant physiology advisor	Credible ground truth, protocol review	Approach ICAR-CPCRI or UAHS Shivamogga scientists
Analog and hardware mentor	Low-noise AFE and field-proof design	Incubator mentor network (IIM Kozhikode LIVE or TBI NIT Calicut)
Hardware intern	PCB bring-up, assembly	Final-year ECE student
Field associate, Karnataka	Installation, weekly checks, farmer liaison	Part-time, local hire during Phase 2

16.4 Next 14 days

- [ ] Correct deck slides 7 to 10 (section 15.1, items 2, 3, 5, 6, 7)
- [ ] Fill the blank application fields with section 14's table and section 5's images
- [ ] Email two arecanut scientists asking for an advisory role and pressure-chamber access
- [ ] Order AD5941 and ADS131M08 evaluation boards; print 2 electrode pods and 1 acoustic sentinel from the .scad files
- [ ] Source 24 arecanut seedlings and a rain-shelter site
- [ ] Book a prior-art search and draft the provisional patent claims
- [ ] Prepare a 2-minute answer to "why will this work on a palm, which has no bark or cambium?" (section 2 caveats + Phase 0 plan)

Sources

- Area and production: Outlook Business; Krishi Jagran 2025; IJIRT 2025; Krishi Jagran 2023
- Prices: acrop.app Shivamogga APMC
- Irrigation practice: DASD; ICAR-CPCRI; KAU
- Stem electrical science: Ganthaler et al. 2019; Nadler et al. 2008; Olive stem electrical study, 2020

- Acoustic emission: Tyree & Sperry 1989; Oletić et al. 2020; Vergeynst et al. 2016; Prometheus protocols
- Competitors: Vivent; Phytech; Farm Progress; Saturas (CORDIS); PitchBook; Fasal (AgFunder); Fasal (Preqin)
- Market context: MarketsandMarkets; IMARC
- Disease and insurance context: Deccan Herald 2024; Deccan Herald 2025
- Grant: GENESIS overview (IC IIT Patna); EIR eligibility example (MUTBI)
- Inputs reviewed: PhytoSense AI pitch deck (10 slides) and MeitY Startup Hub application MEIT-986195 (3 Sep 2026)